

CSS Flexbox CSS Grid @keyframes Positioning & z-index Competitive Arcade

Four arcade cabinets, one hour, zero JavaScript. Predict layouts before they render, solve a CSS crime scene, then build your own neon arcade page using `display: flex`, `display: grid`, `@keyframes`, and `z-index` — written with professional, standards-compliant HTML and CSS.

INSERT COIN ► PRESS START

FORMAT

Teams of 2

3 needs instructor approval

DURATION

60 Minutes

CORE MECHANICS

Predict + Debug + Build

TOTAL POINTS

100 Points

FINAL SUBMISSION DUE

Today · 09/28/2026

4:30 PM hard deadline

Activity Progress: 5 of 9

Overview Arcade Rounds CSS Crime Scene Starter Code Build Challenge Gra



STOP & READ — TEAM RULES FOR EVERY ROUND

ACTIVITY 05 CSS Arcade Night

Light Mode

 **ONE laptop per team.** Only the Driver's machine is open. All other laptops, tablets, and phones stay closed and put away.

 **ALL solutions go in ONE shared Google Doc.** Name it `Activity05-TeamName` and share it with every teammate.

 **Record everything:** Round 1 score + report, every bug diagnosis, fixed CSS, and screenshots, plus Round 3 checkpoint screenshots.

 **No AI tools, search engines, CSS generators, or help from other teams** during timed rounds.

 **Not in the Google Doc = not graded.** Keep version history on — it is your evidence.

 **Submit through iCollege by 4:30 PM today.** One submission per team.

Round 3: Arcade Cabinet Build Challenge

Turn the starter template into a themed, polished, one-page arcade cabinet

30-Minute Team Sprint

00:00 – 05:00 · PLAN

Pick a theme, sketch the layout on paper, assign Driver / Navigator.

05:00 – 25:00 · BUILD

Hit all 6 checkpoints. Test in the browser after every rule.

25:00 – 30:00 ·

POLISH

Validate with W3C, capture checkpoint screenshots, push to GitHub.

✓ The 6 Mandatory CSS Checkpoints

1 Flexbox Layout: A header, nav, or card row using `display: flex` with `justify-content`, `align-items`, and `gap`.

2 CSS Grid Board: At least a 3 × 3 grid built with `grid-template-columns` + `repeat()` (bonus: `grid-template-areas`).

3 Keyframe Animation: `@keyframes` with 0% / 50% / 100% stops using `transform`, plus an `animation-delay` and `forwards` or `both`.

4 Layered Composition: A `position: relative` parent with 2+ `position: absolute` children stacked by explicit `z-index` (think I ♥ NY tee).

5 Micro-Interaction: `:hover` and `:focus-visible` states animated with `transition` on `transform` or `box-shadow`.

6 Professional & Responsive: `:root` custom properties, zero inline styles, a `@media (max-width: 768px)` layout change, and W3C-valid HTML/CSS.

Pick Your Cabinet Theme

Choose one pre-approved theme or pitch a custom concept to your instructor during the planning minutes.



Retro Tic-Tac-Toe Cabinet

A neon arcade machine with a flex marquee header, a 3 × 3 grid board, and a winning line that animates across the board.

Layer idea: a “PLAYER 1 WINS” badge floating above the board with `z-index`.



Midnight Bake-Off

A recipe showdown: flex ingredient cards, a grid of 6+ cookies to vote on, and a “Fresh from the Oven” steam animation.

Layer idea: a ribbon or “Best in Show” stamp positioned over the winning cookie photo.



Math Quest Game Show

A quiz-show stage: a grid of question tiles (Jeopardy style), and answers that fly in with a multi-stage keyframe like the fraction challenge.



Merch Drop Studio

A limited-edition tee shop for your city (I ♥ ATL, I ♥ NYC...): grid product gallery, flex filter bar, animated “DROPS IN 3...2...1” banner.

Layer idea: spotlight and host podium layered over the stage backdrop.

Layer idea: text and a graphic absolutely positioned on a tee drawn with `clip-path` or an image.



Festival Lineup Poster

A concert poster with a grid lineup (headliner spans 3 columns), flex ticket tiers, and a pulsing “SOLD OUT” stamp.

Layer idea: overlapping stage lights and silhouette layers controlled by `z-index`.



Mission Control

A launch dashboard: grid telemetry panels, flex status bar, and a rocket that animates from launchpad to orbit.

Layer idea: rocket, smoke, and HUD overlay stacked in three explicit layers.



Cabinet Launchpad: Build, Test, Document

Add a `Build Challenge` section to your README that names the theme, lists which selector satisfies each checkpoint (e.g. “Checkpoint 2 → `.game-grid`”), and embeds one screenshot per checkpoint. Use the DevTools Flexbox and Grid overlays in your screenshots for extra clarity.



High-Score Bonus Levels (optional, +2 each, max +4)

- **Area Map:** lay out the whole page with named `grid-template-areas`.
- **Motion-Safe:** wrap animations in `@media (prefers-reduced-motion: no-preference)`.
- **Pure-CSS Art:** draw your layered piece entirely with CSS shapes, gradients, or `clip-path`.

In-Class Activity 05: CSS Arcade Night — Flex, Grid, Motion & Layers

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